Scope Honesty in the Collatz-Adjacent Literature: A Running Calibration Against Complete-Proof Claims

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Abstract

A companion note to this survey catalogs twelve claims of a complete proof or disproof of the Collatz conjecture, all of them invalid. This note turns to a different population: sixteen papers, encountered in the same broader literature survey, that make no such claim — results about Collatz-adjacent structures, heuristics, or partial statistics. Read in full and checked computationally where possible, this population behaves very differently from the proof-claim set. The distinguishing axis is not error rate but scope honesty: with no known exception in this sample, these papers correctly label a conjecture a conjecture and a theorem a theorem, in sharp contrast to the systematic overclaiming that defines the companion catalog. Genuine errors do occur — most notably two in a seven-paper program by a single prolific author, one in an otherwise careful standalone result — but they cluster differently: the prolific author's errors are both the same shape (a conjectural or preliminary claim inconsistent with an already-established statement elsewhere in the same text, not an arithmetic slip) and contained to a single section each, while the standalone result's error sits in its titular theorem. This note is explicitly a partial, running document: it consolidates sixteen items reviewed in depth so far, out of a much larger surveyed collection, and is structured to grow as more items are processed.

1 Introduction

A companion note, *Errors in Claimed Complete Proofs (and Disproofs) of the Collatz Conjecture*, catalogs twelve claims of a complete solution to the conjecture, encountered during a broader literature survey, and shows that all twelve fail — most by the same handful of recurring errors. That population is unusual: every item in it explicitly claims to settle the conjecture, a strong claim that invites strong scrutiny.

This note is about the rest of the survey: papers that study Collatz-adjacent structures — accelerated-dynamics reformulations, statistical models of orbit behavior, combinatorial properties of parity sequences, coordinate systems for the state space — without claiming to prove or disprove the conjecture itself. This is a much larger and more heterogeneous population than the proof-claim set, and this note does not attempt to cover all of it: as of this version, it consolidates sixteen items that have been read in full and reviewed in depth, out of a considerably larger set encountered in the survey. It is written as a running document, expected to grow as more of the surveyed collection is processed in later sessions, not as a finished census.

The question this note asks of the sixteen items it covers is not primarily "are they correct" — most are, and correctness in elementary or standard mathematics applied competently is not surprising. The more informative question, and the one that lets this note function as a calibration point against the companion catalog, is scope honesty: does the paper claim only what it establishes? Section 3 examines the largest coherent block in this sample, a seven-paper program by a single author applying one classical ergodic-theory concept per paper to the same underlying quantity. Section 4 covers the remaining nine items, independent of one another and of the program in Section 3. Section 5 draws out the contrast with the companion catalog and states what containment of error does and does not generalize to. Section 6 concludes.

2 Methodology

Each item was read in full and its concrete, checkable claims — identities, closed-form formulas, theorems with an explicit statement — were independently verified computationally wherever they admit a numerical test. Where a paper distinguishes proved results from conjectural ones, that labeling was checked against what the paper's own proof actually establishes, not taken at face value. In the handful of cases where the judgment involved was subtle, the analysis was cross-checked against an independent large-language-model reviewer. As in the companion catalog, every specific claim made below about a named paper traces to this review process, not to a paraphrase of it; full technical detail for each item was recorded during review in internal working notes and is summarized, not reproduced, here.

3 The Ruiz Castillo program

Seven papers by Juan Carlos Ruiz Castillo (ScD., Universidad de San Carlos de Guatemala) appear in this sample, each applying one classical concept from ergodic theory or large-deviations theory — pressure, a residual "geometry," dissipative bounds, a central limit theorem, a transfer operator, Gibbs measures, a variational principle, a large-deviations rate function — to the same elementary quantity: the residual drift $L_k(n) = k\log_2 3 - A_k(n)$, where $A_k(n)$ is the sum of the first k 2-adic valuations along the accelerated Collatz map $U(n) = (3n+1)/2^{v_2(3n+1)}$ — the standard cycle-equation drift already used elsewhere in this research program. None of the seven claims to prove the Collatz conjecture; each says so explicitly. The presentation is heavily repetitive across the series (recurring decorative call-out boxes restating the same conceptual chain, self-citation rates of 75–100% revealing on the order of 19–20 near-identical titles by the same author), but the underlying formal rigor is notably stable: five of the seven papers contain no error we could find, and the two genuine errors found are both contained to a single section and do not affect the papers' own later conclusions.

Geometría Residual [1] applies standard thermodynamic formalism (pressure as a supremum over invariant measures, a Fisher-metric second derivative) to the residual drift, correctly and without claiming to prove Collatz. The same trivial fact — convexity implies a nonnegative second derivative — is presented as five separately named results across the paper, framed by ten decorative call-out boxes restating the same conceptual chain, with no numerical content anywhere (both figures are explicitly labeled "conceptual") and 100% self-citation (13 of 13 references). We computed explicitly, in closed form, the quantity the paper leaves abstract: under the standard i.i.d. model already used in this research program, $\Lambda(t) = t(1-\log_2 3) - \log(2-e^t)$, whose second derivative diverges as $t \to \log 2^-$ — a genuine, if incidental, feature the paper never computes because it never derives $\Lambda(t)$ explicitly.

Dissipative Bounds [2], the same author's second paper in this sample, repeats the pattern: correct elementary identities (an exact affine unrolling of the accelerated map, a universal dissipation bound), no numerical content, and self-citation at 75% (12 of 16 references), with the self-citation list itself revealing roughly a dozen further near-identical titles by the same author on the same "residual decomposition."

Teorema Central del Límite Residual [3] labels its central claim honestly: the title says "Teorema," but the result is stated in the body as "Conjetura 4.2," with five explicit unestablished hypotheses (existence of a residual Gibbs measure, ergodicity, a suitable function space, a spectral gap, positive residual variance), and the paper's own conclusion states plainly that this framework does not prove the Collatz conjecture. We tested the conjecture's observable prediction — asymptotic normality of the residual drift's fluctuations — directly on real Collatz orbits (not the abstract i.i.d. model): variance stabilizes near 2, skewness toward 0, and kurtosis toward 3 as the window length grows, a result the paper itself never tests numerically.

Operador de Transferencia Residual [4] contains this program's first genuine error. Its Proposición 5.3 claims $\lim_{t\to\infty} L_t(1) = 0$; its own proof derives the closed form $L_t(1) = e^{(\log_2 3-1)t}/(1-e^{-t})$ and correctly observes that this "grows exponentially as $t \to \infty$," immediately followed by the proof's closing mark with the "$= 0$" claim never corrected. We confirmed numerically that $L_t(1)$ grows from 2.84 at $t = 1$ to roughly 6.4×10^{50} at $t = 200$, strictly increasing throughout — the opposite of the stated limit. This is a statement-versus-proof inconsistency, not a computational error: the proof itself derives the correct asymptotic and the surrounding text uses that correct asymptotic to motivate the next section's normalization. The error is confined to a preliminary calculation in Section 5; no later, explicitly conjectural result (Sections 6–8, on the restricted transfer operator) depends on it.

Medidas de Gibbs Residuales [5] and **Principio Variacional Residual** [6] return to the error-free pattern: correct restatements of the same core identity, classical convex-analysis apparatus (suprema of affine functions, Legendre–Fenchel duality) correctly applied, and every open result honestly labeled "Conjetura," never "Teorema" or "Proposición."

Grandes Desviaciones Residuales [7] contains this program's second genuine error, structurally similar in kind to the first but located differently: its Definición 3.1 defines a rate function $I_{\mathrm{RC}}(x)$ via the one-sided tail event $\{L_k/k \geq x\}$, and its already-proved Proposición 3.4 (Section 3) correctly shows this one-sided rate function is nondecreasing. Its conjectural Section 7, however — Figura 1, Conjetura 7.3, and Conjetura 7.5, mutually consistent with one another — describes a two-sided, classical Cramér rate function, positive on both sides of its unique zero, contradicting the monotonicity already established in Section 3. We confirmed the contradiction by three independent routes (the standard one-sided large-deviations formula, Monte Carlo simulation, and an exact small-k negative-binomial calculation), all agreeing that the correct one-sided rate is identically zero below the typical drift, not merely at one point. As with the previous case, the error is a conjectural section's inconsistency with an already-proved proposition earlier in the same text, not an arithmetic mistake, and is confined to Section 7; the paper's conclusion (Section 8) does not rely on the specific form of I_{RC}.

4 Nine independent papers

The remaining nine items in this sample share no authorship and little subject matter beyond Collatz-adjacent structures.

Adnan & Dar [8] extend the $3n + 1$ rule to decimals in $(0, 1)$ by defining "parity" via a number's last significant decimal digit, concluding universal divergence. All worked examples in the paper check out arithmetically, but the construction is conceptually ill-posed: "last significant digit" is undefined for any decimal with an infinite expansion (a measure-zero exception in the naive reading, but one that excludes 1/3, 1/7, and every irrational), and even restricted to finite decimals the divergence is a trivial artifact of an unnormalized $+1$ term, not a structural analogue of the real Collatz mechanism (where $3n+1$ is always even for odd n by construction). This is not a proof claim about the original conjecture and does not belong in the companion catalog.

Gilbert [9] constructs an explicit conjugation between the pruned Collatz graph and the nonzero integers, encoding the discarded residue class mod 3 via sign, explicitly disclaiming any proof of the conjecture. Every theorem we checked — the conjugation itself, pruning of multiples of 3, the accelerated map, the parent-count formula, and a connection to OEIS sequence A254046 — holds exactly. No error found.

Seymour (Steiner Sentence Length) [10] derives a matrix of one-step transition probabilities among residues mod 8 (Theorem 2.1), which we confirmed correct, and uses it to derive a claimed closed-form distribution for "Steiner sentence" length, $3^{k-1}/4^k$ (Theorem 5.1), rejecting the naive alternative $(1/2)^k$ via a reported χ^2 test. Our independent reimplementation of the paper's own definitions gives the opposite result: simulation matches $(1/2)^k$, and an exact arithmetic counterexample ($n = 68567$) shows a sentence of length 1 whose entry residue is 7, a case the paper's proof of Theorem 5.1 never accounts for (it only counts sentences whose *first* word already closes, silently missing words with entry 3 or 7 that can close on their first step). We report this with real caution: the paper claims a complete Lean 4/Mathlib formalization with no `sorry`, and we do not have access to that formalization file; it is possible the Lean proof correctly verifies a definition that diverges subtly from the prose statement, in which case the gap would be in the prose-to-formalization translation rather than in the formal proof itself. Based strictly on the definitions as written in the paper, however, both an exact counterexample and a large-scale simulation disagree with the central result.

Seymour (Regular Expression Language) [11], an explicitly labeled working paper by the same author, separates proved theorems from conjectures throughout. Two central results (Proposición 3.1 on Steiner-circuit arithmetic, Teorema 3.6 on run lengths) checked out exactly in large-scale testing. One small, contained error: Corolario 2.2 claims an exit residue mod 24 depends on the parity of an auxiliary variable t; testing shows the claimed value holds only when t is even, and the true residue is the same constant regardless of t's parity, contradicting the paper's own claim that the map $t \mapsto (3t+1)/2$ preserves parity (false in general). This corollary is not listed among the paper's own summary of proved results, and the error does not affect the working paper's central conjecture, which follows almost immediately from the already-verified Theorem 2.1 of the companion paper.

Anthony [12] builds an elaborate formal apparatus (Riemann P-equation, monodromy, hypergeometric identities) around a two-field reformulation of Collatz dynamics via the Möbius coordinate $\Phi(n) = 1/(n+1)$. Every checkable claim — the Möbius reformulation, a theorem excluding indefinite persistence of the $v_2(3n+1) = 1$ regime, and several standard digamma-function identities — checked out exactly. The paper is unusually careful throughout to separate proved theorems from structural analogies (explicitly: "this is an analogy of form, not of kind") and from conditional results whose conditions are not established, and it corrects, within its own text, an earlier erroneous claim from a prior version of the same research line.

Hikawa [13] studies finite combinatorial structures of parity vectors, independent of whether the Collatz conjecture is true or false — a caveat the paper repeats in nearly every section. Both central theorems we tested (an explicit bijection between vector classes, and a rigidity property of a correction term's 2-adic valuation) held exactly by brute force, including the paper's own stated exception case.

Olgac [14] is a three-page note connecting two earlier, uncataloged claims by the same author, one about Collatz and one claiming a complete proof of the Goldbach conjecture (outside this survey's scope). Its only Collatz-specific content is a tautology (root-disjoint branches of the reverse tree, an immediate consequence of unique factorization) rather than a genuine result; the remainder of the note is not stated with enough mathematical precision to evaluate as correct or incorrect.

Williams [15] organizes the positive integers by their 3-smooth factorization and studies Collatz dynamics as a deterministic diagonal flow within that organization. Every theorem we tested — a diagonal-dynamics identity, a theorem on prime-free rows, a bijective partition, an asymptotic count of admissible positions — held exactly. Two small, self-contained errors in OEIS citations were found (one sequence number that does not match the cited property, one off-by-one description), neither affecting any proof in the paper. The paper separates proved theorems from computational observations and explicit open problems throughout, and discloses its use of generative AI (for drafting and formatting, with declared full human verification).

Fu, Liu & Wang [16] propose a Poisson-process mechanism explaining a previously observed Gamma-type statistic for Collatz orbit lengths. Every checkable claim — the expectation and variance of the 2-adic valuation, a closed-form approximation linking orbit length to a starting value, and the algebra of a cycle-closure condition — held exactly. The paper is explicit, correctly and in contrast to the flawed proof attempt reviewed in the companion catalog under the same general argument style, that its asymptotic cycle-closure obstruction is not a proof that no nontrivial cycle exists.

5 Discussion

The contrast that holds across this entire sample, with no known exception, is scope honesty: not one of these sixteen papers dresses a conjecture as a proof of the Collatz conjecture itself, even where the underlying mathematics is elementary and heavily repeated (the Ruiz Castillo program) and even where a central result is probably wrong (Seymour's Steiner sentence paper). This is the axis on which this population differs from the companion catalog's twelve cases, where every single item makes the strong claim and every single item fails to support it. Error rate is not the distinguishing feature here — both populations contain real errors — overclaiming is.

Containment of error is a narrower, true observation, but it holds about the Ruiz Castillo program specifically, not as a property of this population as a whole. Both of that program's genuine errors are the same shape (a conjectural or preliminary claim inconsistent with an already-established statement elsewhere in the same text, not an arithmetic slip) and both are confined to a single section, leaving the paper's later, explicitly conjectural material untouched. Seymour's Steiner sentence paper is the sharp counterpoint: its error sits in the titular Theorem 5.1, the paper's central claimed result, not a peripheral lemma. The interesting fact is not that errors in this literature are contained — they are not, in general — but that the most prolific single-author program in this sample errs in a consistently contained way, while the sharpest standalone result in the sample errs at its center. That difference is a fact about how particular authors err, not a generalization about the population.

We note, finally, that this chapter's coverage is partial by construction: sixteen items reviewed in depth here, drawn from a considerably larger surveyed collection that includes both further unprocessed Collatz-adjacent papers and a separate body of established reference literature (Tao's almost-all-orbits theorem, the Kontorovich–Lagarias stochastic model, and similar) that is cited throughout this research program but was never a target of this kind of review. Future revisions of this note are expected to add further items as they are processed; the structure above — one section per coherent author or program, individual paragraphs per independent item — is meant to accommodate that growth without requiring reorganization.

6 Conclusion

Sixteen Collatz-adjacent papers, none claiming to prove or disprove the conjecture, reviewed in full: nine contain no error we could find; four contain a small, contained error (two in the Ruiz Castillo program, one in a Seymour working paper, and one, a pair of citation-only slips, in Williams); one — Seymour's Steiner sentence paper — has a likely error in its central, titular theorem rather than a peripheral one; and two make no claim evaluable as straightforwardly correct or incorrect (Adnan & Dar's construction is conceptually ill-defined; Olgac's only Collatz-specific content is tautological). Across all sixteen, the theorem-versus-conjecture distinction is maintained with a rigor this survey's companion catalog of proof claims never shows. That contrast — scope honesty, not error rate — is this note's calibration point, and it is expected to sharpen, not change in kind, as more of the surveyed collection is added in later revisions.

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